

CO2 - AT3

1. Image Enhancement in the Spatial Domain.

Image enhancement is the process of improving the visual quality of an image so that it becomes easier to analyze. In the spatial domain, enhancement is done by directly changing the pixel values of the image.

1) Gray-level Transformation.

Gray-level transformation modifies the intensity values of pixels.

- Negative Transformation:

It highlights white details on dark backgrounds.

- Log transformation: Brightens dark regions

- Gamma transformation: Adjusts the overall brightness and contrast

Uses: Medical images, satellite images.

2) Histogram processing:

A histogram shows the distribution of pixel intensities in an image.

Histogram equalization: It improves the contrast by spreading pixel values over the full intensity range.

Uses: Low contrast image and image preprocessing.

3) Spatial Filtering.

Spatial filtering uses a small mask (kernel) to process the image.

Smoothing filters: It removes noise and blur unwanted details.

Sharpening filters: Enhance edges and fine details

Conclusion:

Gray-level transformation improve brightness, histogram processing improves contrast and spatial filtering removes noise or sharpens edges.

2) Image Smoothing and Image Sharpening.

a)

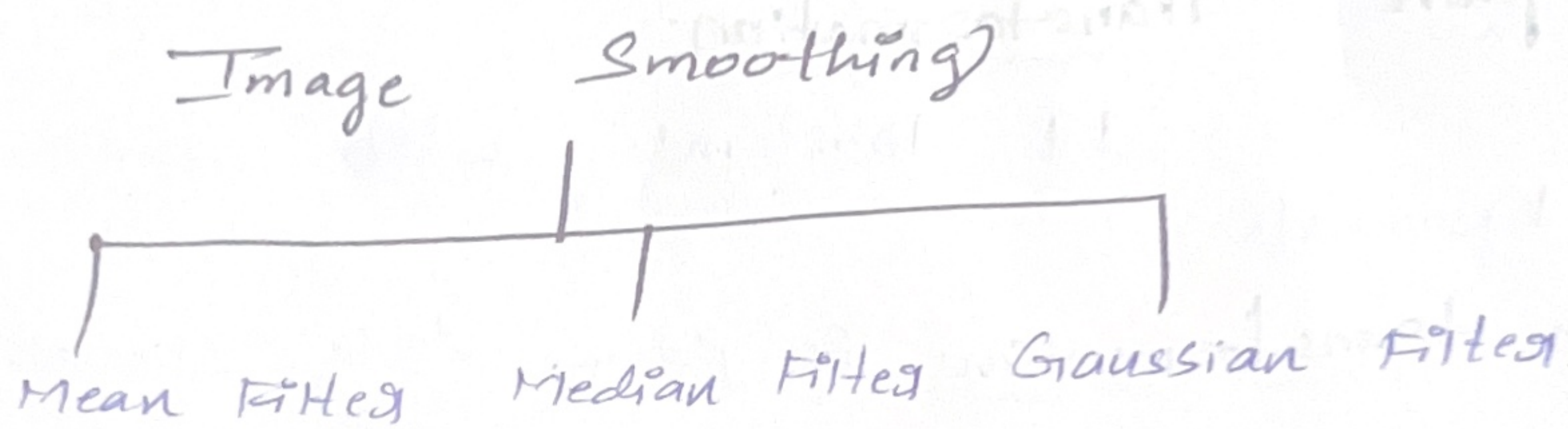


Image smoothing and sharpening are common in image enhancement techniques.

Image smoothing reduces noise and small intensity variations.

Working principle:

Each pixel is replaced by the average or median value of the neighbouring pixel values.

Advantages:

- It removes noise
- produces smoother images
- useful before segmentation.

Applications:

Medical images, CCTV image and image preprocessing

b) Image Sharpening.

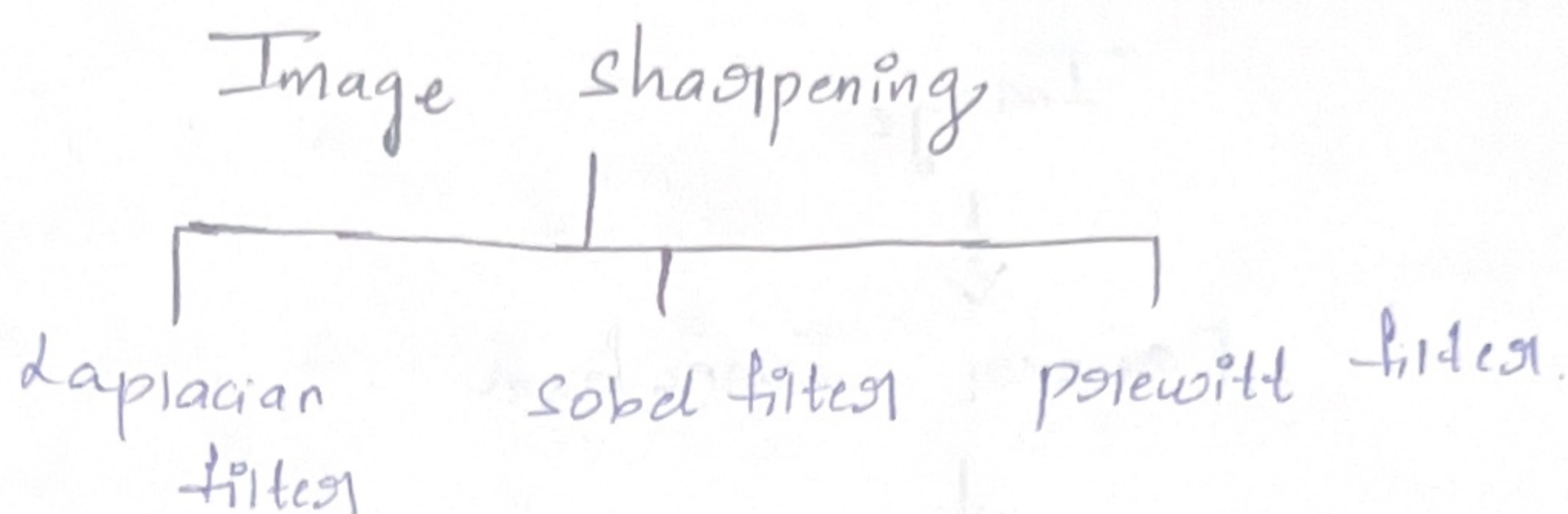


Image sharpening increases the edges and makes fine image.

Working Principle:

It enhances rapid change in pixel intensity using high-pass filter.

Advantages:

- Highlights edges
- Improve image details
- Helps in object recognition.

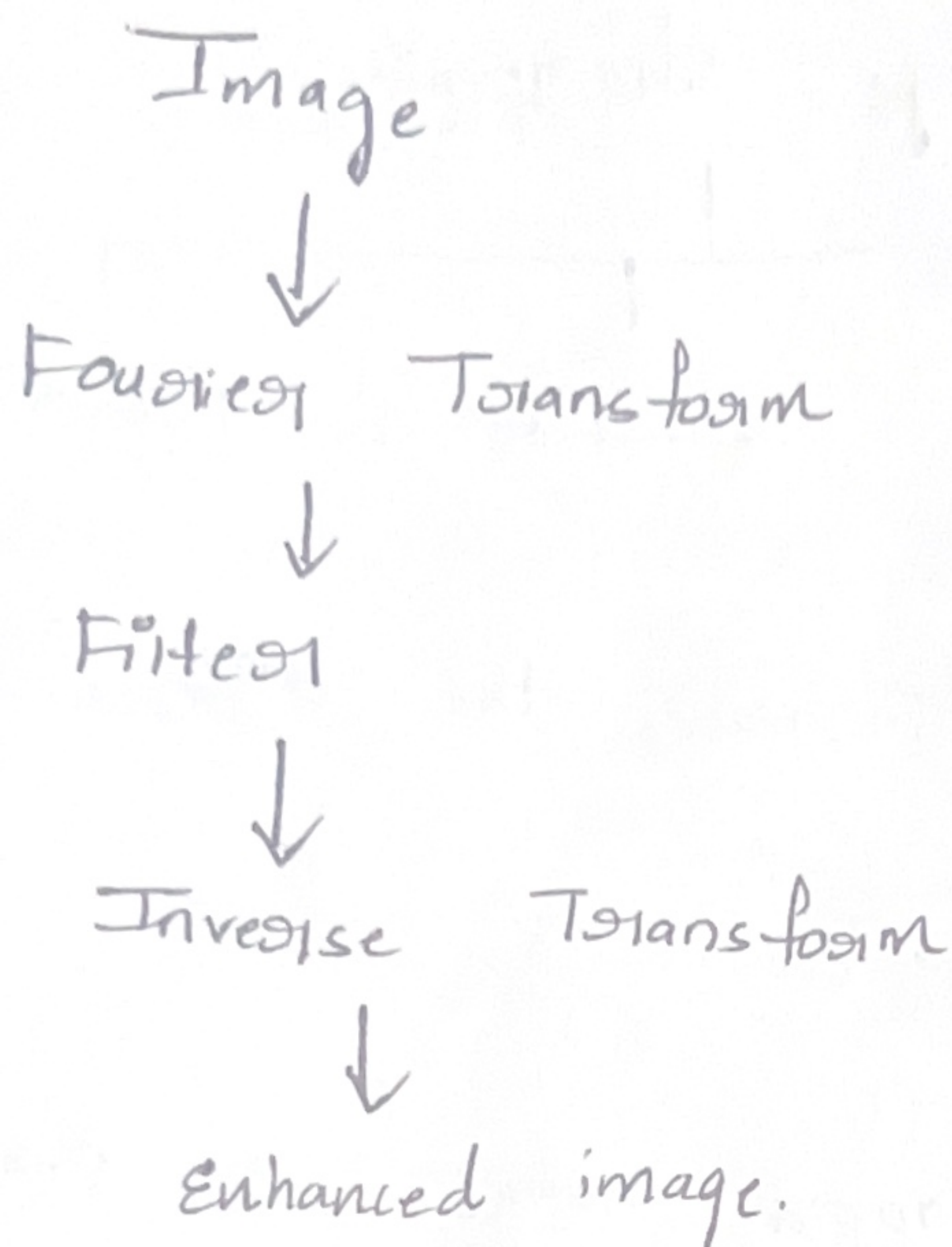
~~Applications:~~

- ~~Fingerprint recognition~~
- Face detection
- Industrial inspection.

Conclusion: Smoothing is used for noisy images, while sharpening is used to improve image details.

Image Enhancement in the Frequency Domain.

Frequency-domain enhancement improves an image by processing its frequency components using the Fourier Transform (FT).



Low-Pass Filter (LPF):

A low pass filter allows low-frequency components and removes high-frequency components.

Effects:

- Removes noise
- Smooths image.
- Reduces sharp edges

High-pass Filter:

A high pass filter allows high-frequency components and removes low-frequency components.

Effects:

- Enhances edges
- Sharpens image
- Highlights fine details

4) Thresholding

Separates the object from the background using intensity values.

Advantages:

- Easy to implement
- Fast.

Applications:

Document Scanning and OCR.

5) Region - Based Segmentation

Groups neighboring pixels with similar properties

Advantages:

- produces complete regions
- More accurate segmentation.

Conclusion:

For most applications, edge detection combined with region based Segmentation gives better object boundaries and accurate Segmentation.

5) Edge Detection, Edge Linking and Boundary Detection

These techniques are important for identifying objects and detecting in images.

Edge Detection:

edge detection identifies sudden changes in pixel intensity

Working Principle:

It uses operators such as Sobel, Prewitt, Roberts and Canny to detect edges.

continuous edges ^{connects} broken edge segments to form working principle:

It joins nearby edge pixel using local information of algorithm like Hough transform.

Importance:

- Produces continuous boundaries
- Improves Segmentation accuracy.

Boundary detection:

Boundary detection identifies the complete outer borders of an object.

working principle:

It traces the connected edges or segmented regions to form a closed boundary.

Importance:

- Measures object size and shape
- Helps in defect detection
- Improves object recognition.

Conclusion:

edge detection finds edges, edge linking connects them, and boundary detection forms the complete object outline. Together, these techniques improve object recognition and defect detection in computer vision.